Transition Analysis Toolkit
A Reproducible Framework for Analysing Student Transition Pathways

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Version 0.9

Draft for Review

August 5, 2026

Abstract

Student transition into first-year university mathematics remains a significant challenge for higher education institutions, with differences in prior mathematical preparation strongly influencing subsequent academic success. This study investigates factors associated with successful transition into a first-stage undergraduate mathematics subject using the Transition Analysis Toolkit, a reproducible analytical framework that integrates statistical analysis, interpretable machine learning, independent verification, and automated publication generation. Decision tree and random forest models were developed and compared using a reproducible analytical workflow to evaluate predictive performance and the relative importance of educational predictors. Across all evaluated models, secondary mathematics preparation consistently emerged as the strongest predictor of academic success, followed by composite risk measures and preparatory mathematics. Beyond the empirical findings, the principal contribution of this research is the development of a reproducible workflow that links data preparation, statistical modelling, machine learning, independent verification, automated reporting, and manuscript generation within an integrated framework for educational research. The framework demonstrates how reproducible analytical workflows can support interpretable educational data analytics while ensuring consistency between analytical outputs and published results. It provides a reusable foundation for future investigations of student transition, progression, and academic success across educational contexts and supports evidence-informed institutional decision-making.

Keywords: student transition; university mathematics; mathematical preparedness; interpretable machine learning; reproducible research

1 Introduction

The transition from secondary school to first-year university mathematics presents a well-documented challenge for many students entering science, engineering, and related disciplines [Clark and Lovric, 2009]. Differences in prior mathematical preparation, curriculum pathways, and academic readiness contribute to substantial variation in students' achievement and persistence during the first year of university study [Hourigan and O'Donoghue, 2007]. For universities, understanding these factors is important for informing curriculum design, identifying students who may benefit from targeted academic support, and developing evidence-based transition initiatives [Thomas, 2008]. This study investigates the factors associated with successful transition into first-year university mathematics with the aim of providing evidence to inform institutional decision-making and support student success.

Over the past two decades, considerable research has examined the factors associated with success in first-year university mathematics, including prior academic achievement, secondary school mathematics preparation, demographic characteristics, and measures of academic preparedness [Clark and Lovric, 2009, England et al., 2021]. More recently, advances in educational data analytics and machine learning have enabled increasingly sophisticated approaches to modelling student performance [Koedinger et al., 2015, Heffernan et al., 2023]. While these approaches have improved predictive capability, there remains a need for analyses that not only identify influential predictors but also provide interpretable evidence of their relative importance in ways that are meaningful for educational decision-making. This study addresses that need through a transparent and reproducible analytical framework that combines predictive modelling with educational interpretation to investigate the factors associated with successful transition into first-year university mathematics.

This paper presents a reproducible framework for analysing student transition pathways into first-stage undergraduate mathematics through an integrated analytical workflow combining statistical analysis, interpretable machine learning, independent verification, and automated reporting. As a case study, the framework is applied to investigate factors associated with successful transition into a first-stage undergraduate mathematics subject using multiple tuned decision tree and random forest models. Rather than advocating a single predictive model, the study emphasises consistency of interpretation across modelling approaches and demonstrates how transparent, reproducible analytical workflows can support educational research. The principal contribution is therefore not a new predictive algorithm, but a reusable research framework that integrates data preparation, model development, independent verification, interpretation, and publication-ready reporting within a single reproducible pipeline. The overall research workflow is described in §2.1.

1.1 Research Questions

1. Which factors are the strongest predictors of success in first-year university mathematics?
2. How consistent are predictor importance rankings across multiple machine learning models?
3. What educational insights emerge from interpreting the tuned models?

2 Methodology

We adopted a reproducible analytical workflow to investigate factors associated with successful transition into first-year university mathematics. The workflow was designed to ensure that each stage of the analysis, from data preparation through statistical modelling, independent verification, and report generation, could be reproduced from the original research dataset using an automated build process. This approach emphasises transparency, repeatability, and traceability, allowing both analytical results and publication outputs to be generated consistently from the same underlying evidence.

2.1 Research Workflow

We followed a reproducible analytical workflow consisting of data preparation, statistical analysis, machine learning model development, independent verification using R, and automated generation of publication-ready tables, figures, and manuscript components. Each stage formed part of an integrated build process, ensuring that analytical outputs could be reproduced directly from the underlying research dataset.

Figure 1 summarises the reproducible analytical workflow used in this study, from preparation of the approved research dataset through statistical analysis, machine learning, independent verification, and automated publication outputs. The analytical dataset used throughout the study is described in §2.2.

2.2 Research Dataset

The study used de-identified institutional data from students enrolled in a first-stage undergraduate mathematics subject offered during the Autumn 2026 teaching session. The dataset was prepared specifically for research purposes using an approved de-identification process that removed all personally identifying information and replaced student identifiers with research identifiers, referred to as coding. The resulting research ready dataset retained the academic, demographic, and derived analytical variables required for the statistical and machine learning analyses undertaken in this study. The variables included in the analytical dataset are described in §2.3.

2.3 Variables

The analytical dataset included demographic, prior educational, and academic performance variables used to investigate successful transition into first-stage university mathematics. Variables were derived from de-identified institutional records following data preparation and were grouped into categories representing secondary mathematics preparation, previous university mathematics performance, preparatory mathematics experience, student characteristics, current academic outcome, and the institutional composite risk index. Table 1 summarises the variables included in the analysis.

The variables were selected to address the research questions by representing students' prior mathematical preparation, previous university mathematics experience, and institutional measures available before or during enrolment. Together, these variables formed the basis for the statistical analyses described in §2.4.

Institutional Source Data

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Data Preparation

↓

Research Dataset

↓

Statistical Analysis | **Machine Learning Analysis**

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Independent Verification (R)

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Tables • Figures • Generated LaTeX Values

↓

Publication-ready Paper

Figure 1: Reproducible analytical workflow implemented by the Transition Analysis Toolkit. Institutional data are transformed into an approved de-identified research dataset before parallel statistical and machine learning analyses are independently verified in R. Automated generation of tables, figures, LaTeX variables, and manuscript content ensures consistency between the analytical outputs and the final publication.

Table 1: Summary of variables included in the analytical dataset.

Variable	Description
Secondary mathematics preparation	Highest level of secondary school mathematics completed
Secondary school mark	Overall secondary mathematics achievement
Previous university mathematics	Previous university mathematics enrolment and performance
Preparatory mathematics	Previous enrolment and performance in the preparatory mathematics subject
Student characteristics	Demographic characteristics included in the analysis
Current academic outcome	Pass or fail outcome for the study subject
Composite risk index	Institutional composite measure of student academic risk

2.4 Statistical Analysis

We conducted statistical analyses to summarise the characteristics of the research dataset and to examine relationships between predictor variables and academic outcomes. Descriptive statistics were generated to provide an overview of the cohort, while inferential analyses supported comparison of predictor variables prior to the development of machine learning models [Field, 2013]. These analyses established the statistical context for the machine learning models described in §2.5.

2.5 Machine Learning Models

We developed machine learning models to investigate the relative importance of factors associated with successful transition into first-year university mathematics. Decision tree and random forest classifiers were selected because they provide competitive predictive performance while also supporting interpretable measures of feature importance [Breiman et al., 1984, Breiman, 2001]. Hyperparameter tuning and model evaluation were undertaken using reproducible workflows to ensure consistent comparison across modelling approaches.

2.6 Independent Verification

We implemented an independent verification pipeline using the R statistical computing environment [R Core Team, 2026, Hinz and Yee, 2018, Hinz, 2021] to strengthen confidence in the analytical results. Key statistical summaries and model outputs were reproduced independently of the primary Python implementation to confirm the consistency and reliability of the analytical workflow. This independent verification formed an additional quality assurance stage prior to the generation of publication outputs.

2.7 Automated Reporting

The final stage of the analytical workflow automatically generated publication-ready tables, figures, model summaries, and manuscript components directly from the verified analytical outputs. By integrating analysis and reporting within a single reproducible workflow, we reduced the potential for manual transcription errors while ensuring that all reported results remained synchronised with the underlying research data and analyses.

The methodology described above established a transparent and reproducible analytical framework for investigating factors associated with successful transition into first-year university mathematics. The following section presents the results obtained using this analytical framework.

3 Results

This section presents the findings generated by the reproducible analytical workflow described in the preceding methodology. Results are presented in four stages. First, the predictive performance of the machine learning models is evaluated. Second, the relative importance of predictor variables is examined. Third, the consistency of feature rankings across modelling approaches is assessed. Finally, the educational significance of these findings is interpreted in the context of student transition into first-year university mathematics.

3.1 Model Performance

The predictive performance of the tuned Decision Tree and Random Forest models was evaluated using cross-validated and independent test-set ROC AUC. The comparison provides an assessment of each model's ability to distinguish between students who successfully completed the subject and those who did not. Results from the tuned models are summarised in Table 2.

Table 2: ROC AUC comparison for cross-validated Decision Tree and Random Forest models.

Model	CV ROC AUC	Test ROC AUC	Test Rows
Decision Tree — With Risk	0.7072	0.7664	295
Random Forest — Without Risk	0.7101	0.7603	295
Random Forest — With Risk	0.7095	0.7599	295
Decision Tree — Without Risk	0.6994	0.7557	295

The ROC analysis demonstrates that all four tuned models achieved moderate discriminatory performance on the independent test dataset, with ROC AUC values ranging from approximately 0.756 to 0.766. Although the observed differences between models were relatively small, the Decision Tree — With Risk achieved the highest test ROC AUC of 0.766. These findings indicate that each modelling approach provides meaningful predictive capability, while suggesting that the inclusion of the composite risk measure provides a modest improvement in identifying students at greater risk of unsuccessful outcomes.

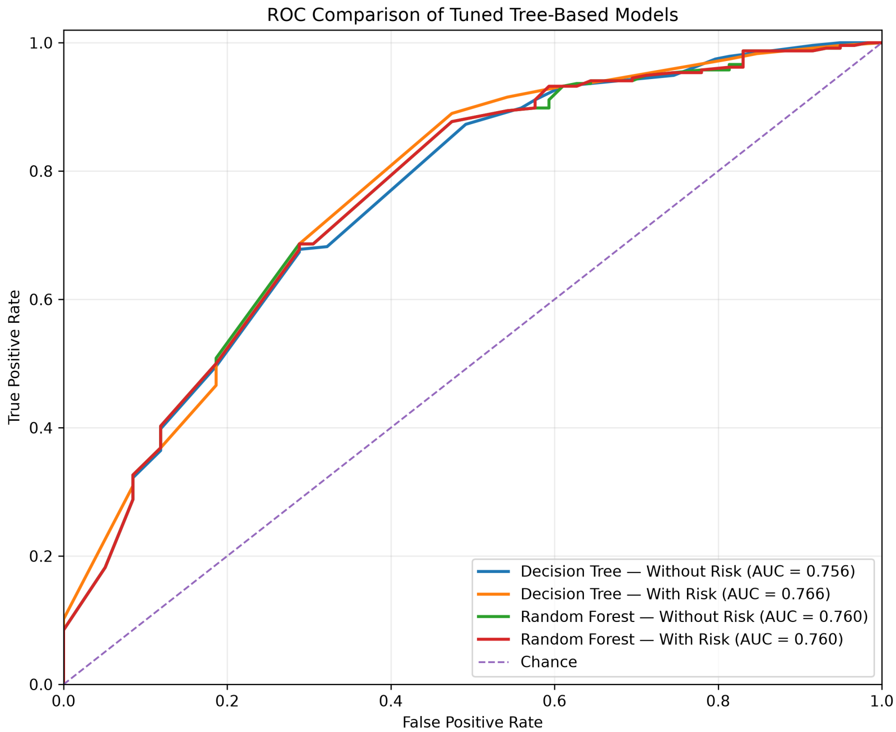

Figure 2: Receiver Operating Characteristic (ROC) curves for the tuned Decision Tree and Random Forest models evaluated on the independent test dataset. Curves closer to the upper-left corner indicate stronger discriminatory performance, while the area under each curve (AUC) summarises predictive performance across all classification thresholds.

3.2 Feature Importance

Rank	Research theme	Mean importance
1	Secondary mathematics preparation	0.2755
2	Composite risk measure	0.2381
3	Preparatory mathematics	0.2320
4	Previous university performance	0.1090
5	Secondary school achievement	0.1058
6	Student characteristics	0.0395

3.3 Feature Rankings

Rank	Feature	Models	Mean importance
1	Composite risk index	2	0.4762
2	Prior preparatory mathematics	4	0.1694
3	Secondary mathematics: Other	4	0.1019
4	Secondary mathematics: Standard	4	0.0950
5	Previous attempt at current subject	4	0.0693
6	Previous preparatory mathematics pass	4	0.0532
7	Previous failure in current subject	4	0.0397
8	International student	4	0.0395
9	Secondary school mark below 70	4	0.0394
10	Secondary mathematics: Extension 1	4	0.0348
11	Secondary school mark 90 or above	4	0.0288
12	Secondary mathematics: Extension 2	4	0.0257
13	Secondary school mark 80–89	4	0.0228
14	Secondary mathematics: Advanced	4	0.0182
15	Secondary school mark 70–79	4	0.0148
16	Previous preparatory mathematics failure	4	0.0094

3.4 Model Interpretation

Across the tuned tree-based models, secondary mathematics preparation emerged as the most influential predictor group (mean importance = 0.276). This was followed by the composite risk measure (0.238) and preparatory mathematics (0.232). Together, these three groups accounted for 74.6% of the average feature importance across the tuned models. Secondary mathematics preparation remained influential across all four models, whereas the composite risk measure contributed strongly but appeared in two of the four models. In contrast, student characteristics made the smallest average contribution. This suggests that prior secondary mathematics preparation was the strongest and most consistent predictor of success across the tuned models.

4 Discussion

The results of this study provide new insights into the factors associated with successful transition into first-stage university mathematics. By combining reproducible analytical methods with interpretable machine learning models, the analysis extends beyond prediction alone to examine the relative importance and consistency of the variables associated with student success. This discussion considers these findings in the context of previous research, reflects on their educational significance, and identifies implications for both institutional practice and future research.

Secondary mathematics preparation emerged as the strongest and most influential predictor group across the tuned models, with a mean feature importance of 0.276. Composite risk (0.238) and preparatory mathematics (0.232) were the next most influential predictor groups. Collectively, these three groups accounted for 74.6% of the average feature importance, indicating that measures of prior mathematical preparation dominated the predictive performance of the evaluated models. Furthermore, the consistency of the feature rankings presented in Section 3.3 showed that secondary mathematics preparation remained influential across all four tuned models, whereas the composite risk measure contributed strongly but appeared in only two models. Together, these findings provide robust evidence that students' secondary mathematical preparation was the strongest and most consistent predictor of success in this cohort.

These findings suggest that successful transition into first-stage university mathematics is influenced primarily by students' existing mathematical preparedness rather than by demographic or administrative characteristics alone. The consistency of secondary mathematics preparation across multiple modelling approaches reinforces the importance of prior mathematical knowledge as a foundation for university success. At the same time, the contribution of preparatory mathematics and the composite risk measure indicates that institutional interventions can play a meaningful role in identifying and supporting students whose prior preparation may place them at greater academic risk.

Relationship to Previous Research

Previous research has consistently demonstrated that students' prior mathematical preparation influences their transition to university mathematics [Hourigan and O'Donoghue, 2007, Wood, 2001, England et al., 2021]. This includes previous work undertaken by the author [Stanley, 2025]. The findings suggest that institutions should continue to place strong emphasis on identifying students' mathematical preparedness before or at the commencement of university study [Reddy et al., 2025]. Rather than contradicting existing evidence, the results strengthen current understanding by demonstrating that these relationships remain evident across multiple analytical approaches [Hourigan and O'Donoghue, 2007, Wood, 2001, England et al., 2021].

Contribution of the Present Study

The principal contribution of this study is not the identification of prior mathematical preparation as an important predictor of student success, since this has been consistently reported in previous transition research. Rather, the contribution lies in the development and application of a transparent, reproducible analytical workflow that integrates statistical analysis, interpretable machine learning, independent verification, and automated research reporting. This framework enables educational researchers to investigate transition outcomes using methods that are both explainable and reproducible, while maintaining a clear connection between quantitative evidence and educational interpretation.

Educational Implications

The findings suggest that institutions should continue to place strong emphasis on identifying students' mathematical preparedness before or at the commencement of university study. Rather than viewing prior preparation as a fixed limitation, the results support the use of early diagnostic measures and targeted preparatory programs to identify students who may benefit from additional support. The prominence of preparatory mathematics and the composite risk measure indicates that institutional interventions can complement students' existing knowledge by providing structured pathways that improve their likelihood of success.

Limitations and Future Research

The present study should be interpreted within the context of several limitations. The analysis was conducted using data from a single first-stage university mathematics subject and therefore reflects the characteristics of one institutional context rather than the broader higher education sector. Although the analytical workflow was intentionally designed to be reproducible and transferable, the specific predictive relationships identified in this study may vary across institutions, disciplines, and student cohorts. Future research should apply the framework to additional subjects, longitudinal datasets, and diverse institutional settings to evaluate its broader generalisability while continuing to refine interpretable approaches for educational learning analytics.

5 Conclusion

This study has demonstrated that a transparent and reproducible analytical workflow can be successfully applied to the investigation of student transition into first-stage university mathematics. By integrating statistical analysis, machine learning, independent verification, and automated publication generation within a single framework, the Transition Analysis Toolkit provides a robust foundation for both educational research and evidence-informed institutional decision making.

The analytical results consistently demonstrated that prior mathematical preparation was the strongest predictor of success in first-stage university mathematics. Across both statistical models and interpretable machine learning approaches, secondary mathematics preparation emerged as the most influential factor, while demographic characteristics contributed comparatively little to predictive performance. These findings reinforce previous research highlighting the importance of discipline-specific preparation during the transition to university and demonstrate that interpretable machine learning can complement traditional statistical methods without sacrificing transparency [Hourigan and O'Donoghue, 2007, Wood, 2001, England et al., 2021].

The study also demonstrates the value of an integrated and reproducible analytical workflow for educational research. Statistical modelling, machine learning, independent verification, automated reporting, and publication-ready document generation were combined within a single research framework, ensuring that all reported results were derived directly from the underlying analysis. This approach reduces opportunities for transcription errors, improves transparency, and provides a reusable foundation for future studies investigating student transition, progression, and academic success across comparable educational contexts.

The framework developed in this study provides a foundation for future investigations of student transition using larger cohorts, additional academic disciplines, and longitudinal datasets. While the present case study focused on a single first-year mathematics subject, the methodology is readily extensible to other educational contexts where transparent interpretation of predictive models is essential. Future work will investigate broader institutional datasets, evaluate additional modelling approaches, and further develop the automated research workflow to support reproducible educational analytics and evidence-informed decision making.

Ultimately, this research demonstrates that reproducible educational analytics can extend beyond predictive modelling to encompass the complete research lifecycle. By integrating data preparation, statistical analysis, machine learning, independent verification, automated reporting, and manuscript generation within a unified workflow, the Transition Analysis Toolkit provides a transparent framework that supports both rigorous educational research and practical institutional application. As educational datasets continue to increase in scale and complexity, approaches that combine analytical performance with reproducibility and interpretability will become increasingly important for advancing evidence-informed practice.

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A Machine Learning Primer

This study is accompanied by a standalone machine learning primer that introduces the statistical and machine learning concepts used throughout the analysis, including decision trees, random forests, receiver operating characteristic (ROC) curves, feature importance, and the reproducible computational workflow.

The primer is maintained separately from this paper so that it can evolve as an independent educational resource and be reused across future research and teaching projects.

Repository:

`https://github.com/jasonnstanley/transition_analysis_ml_primer`